

WHEN IS THE RING OF INTEGER-VALUED POLYNOMIALS OVER A GROUP RING A PRÜFER DOMAIN?

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ABSTRACT. In their study of the ring of integer-valued polynomials in non-commutative algebra, Peruginelli and Werner characterized the algebras for which this ring is a Prüfer domain. Here, we apply their results to the case of group algebras.

1. INTRODUCTION

In all the paper, D denotes a commutative integral domain distinct from its quotient field K .

Definition 1.1 (only for this paper). A D -algebra $\mathcal{A}$ is said to be a *nice D -algebra* if it is unitary, is finitely generated and torsion-free as a D -module and is such that $\mathcal{A} \cap K = D$.

In the sequel, $\mathcal{A}$ always denotes a nice D -algebra. Let $\mathcal{B} = K \otimes_D \mathcal{A}$ and $m = \dim_K \mathcal{B}$. With these hypotheses, we may assume that $\mathcal{B}$ contains K and $\mathcal{A}$.

Examples 1.2. The following rings are nice D -algebras.

- a) $\mathcal{M}_n(D)$ the ring of square matrices of size n with entries in D .
- b) $\mathbb{H}_D$ the ring of Hamilton quaternions with coefficients in D .
- c) $D[G]$ the group ring where G is a finite group.

In a recent paper, Peruginelli and Werner characterized the case where the following ring of *integer-valued polynomials on $\mathcal{A}$* is a Prüfer domain:

$$\text{Int}_K(\mathcal{A}) = \{f(X) \in K[X] \mid f(\mathcal{A}) \subseteq \mathcal{A}\}.$$

Theorem 1.3. [5, Theorem 1.7 (1)] *Assume that $\mathcal{A}$ is a nice D -algebra. Then, the ring $\text{Int}_K(\mathcal{A})$ is a Prüfer domain if and only if D itself is a Prüfer domain and $\mathcal{A} = \mathcal{A}'$ where $\mathcal{A}'$ is the set formed by the elements of $\mathcal{B}$ which are integral over D .*

Applying their results to the first two examples of nice algebras, they show that a matrix algebra $\mathcal{M}_n(D)$ where $n \geq 2$ never leads to a Prüfer domain ([2, § 4] or [4, Corollary 3.4]), whereas the ring $\text{Int}_{\mathbb{Q}}(\mathbb{H}_{\mathbb{Z}_{(2)}})$ is a Prüfer domain where $\mathbb{H}_{\mathbb{Z}_{(2)}}$ denotes the Hurwitz quaternion ring on $\mathbb{Z}_{(2)}$ [5, Theorem 5.4], that is,

$$\mathbb{H}_{\mathbb{Z}_{(2)}} = \left\{ a_0 + a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k} \mid a_i \in \mathbb{Z}_{(2)} \ \forall i \text{ or } a_i \in \mathbb{Z}_{(2)} + \frac{1}{2} \ \forall i \right\}.$$

When the Jacobson radical $J(D)$ of D is equal to (0) and $\text{Int}_K(\mathcal{A})$ is a Prüfer domain, then the D -algebra $\mathcal{A}$ is necessarily commutative and we have a more precise result:

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Theorem 1.4. [5, Theorem 1.7 (2)] *Assume that the Jacobson radical $J(D)$ of D is (0) and that $\text{Int}(D)$ is a Prüfer domain. Then, $\text{Int}_K(\mathcal{A})$ is a Prüfer domain if and only if*

$$\mathcal{A} \simeq \prod_{i=1}^t \mathcal{A}_i$$

where $\mathcal{A}_i$ is the integral closure of D in a finite extension F_i of K . In this case, each $\mathcal{A}_i$ is a commutative integral domain such that $\text{Int}_{F_i}(\mathcal{A}_i)$ is a Prüfer domain.

2. GROUP RINGS

Let G be a group and let $D[G]$ be the D -algebra

$$D[G] = \left\{ \sum_{g \in G} a_g g \mid a_g \in D \text{ almost all zero} \right\}.$$

It is a free D -module with basis $\{g\}_{g \in G}$ and a multiplication defined by

$$\left(\sum_{h \in G} a_h h \right) \times \left(\sum_{k \in G} b_k k \right) = \sum_{g \in G} \left(\sum_{hk=g} a_h b_k \right) g.$$

The group ring $D[G]$ is a nice D -algebra if and only if G is a finite group. Thus, from now on we assume that G is a finite group. Of course, if $\mathcal{A} = D[G]$, then $\mathcal{B} = K[G]$ and

$$m = \dim_K K[G] = \text{rk}_D(D[G]) = |G|.$$

By Theorem 1.4, if $J(D) = (0)$ and if $\text{Int}_K(D[G])$ is a Prüfer domain, then $D[G]$ is commutative. Clearly, $D[G]$ is commutative if and only if G is abelian.

The purpose of this text is to characterise the finite groups G such that

$$\text{Int}_K(\mathcal{O}_K[G]) = \{f \in K[X] \mid f(\mathcal{O}_K[G]) \subseteq \mathcal{O}_K[G]\}$$

is a Prüfer domain, where K denotes an algebraic number field and $\mathcal{O}_K$ its ring of integers. As previously said, G has to be abelian. We are going to prove that this necessary condition is also sufficient.

Example 2.1. Let C_p be a cyclic group whose order is a prime number p and let $\zeta_p \in \mathbb{C}$ be a p -th primitive root of unity. We have the following ring isomorphisms:

$$\mathbb{Z}[C_p] \simeq \mathbb{Z}[X]/(X^p - 1) \simeq \mathbb{Z}[X]/(X - 1) \times \mathbb{Z}[X]/(1 + X + \dots + X^{p-1}) \simeq \mathbb{Z} \times \mathbb{Z}[\zeta_m].$$

Consequently,

$$\text{Int}_{\mathbb{Q}}(\mathbb{Z}[C_p]) = \text{Int}_{\mathbb{Q}}(\mathbb{Z}) \cap \text{Int}_{\mathbb{Q}}(\mathbb{Z}[\zeta_m]) = \text{Int}_{\mathbb{Q}}(\mathbb{Z}[\zeta_m]).$$

But we know that this last ring is a Prüfer domain (cf. [2] or, for instance, see [1, Corollary VIII.60]).

In fact, this example may be easily generalized thanks to the following general result that we recall:

Theorem 2.2. [3, Theorem 1] *If G is a finite abelian group of order n and if the characteristic of the field K does not divide n , the group algebra $K[G]$ is isomorphic to a product of rings of the form*

$$K[G] \simeq \prod_{d|n} (K(\zeta_d))^{a_d}$$

where $a_d \in \mathbb{N}$ and ζ_d is a d -th primitive root of unity. More precisely,

$$a_d = \frac{\#\{g \in G \mid \text{ord}(g) = d\}}{[K(\zeta_d) : K]}.$$

Corollary 2.3. *Assume that K is an algebraic number field, with ring of integers $\mathcal{O}_K$, and that G is a finite group with exponent m . The ring $\text{Int}_K(\mathcal{O}_K[G])$ is a Prüfer domain if and only if G is commutative. In that case,*

$$\text{Int}_K(\mathcal{O}_K[G]) = \text{Int}_K(\mathcal{O}_K[\zeta_m]).$$

Proof. By Theorem 1.4, if $\text{Int}_K(\mathcal{O}_K[G])$ is a Prüfer domain, then G has to be commutative. Conversely, assume that G is commutative, then it follows from Theorem 2.2 that, $\mathcal{O}_K[G] = \prod_{d \mid m} (\mathcal{O}_K[\zeta_d])^{a_d}$. Consequently:

$$\text{Int}_K(\mathcal{O}_K[G]) = \cap_{d \mid m} \text{Int}_K(\mathcal{O}_K[\zeta_d]) = \text{Int}_K(\mathcal{O}_K[\zeta_m]) = \text{Int}_K(\mathcal{O}_K(\zeta_m)).$$

Finally, (by [2, Theorem 3.7] or [1, Corollary VIII.60]), $\text{Int}_K(\mathcal{O}_K[G])$ is a Prüfer domain. $\square$

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